

Algebra 1
Unit 5
Lesson 1 Practice

Name Answer Key
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1. When solving an equation, the goal is to isolate the variable.

Two students, Josephine and Jackie, were solving the equation $39 - 8n = -8(3 + 4n) + 3n$. Consider their work is shown below for questions 2–5.

Josephine's Work	Jackie's Work
$ \begin{array}{r} 39 - 8n = -8(3 + 4n) + 3n \\ \underline{-3n} \qquad \qquad \underline{-3n} \\ 39 - 11n = -8(3 + 4n) \\ 39 - 11n = -24 - 32n \\ \underline{+32n} \qquad \qquad \underline{+32n} \\ 39 + 21n = -24 \\ \underline{-39} \qquad \qquad \underline{-39} \\ 21n = -63 \\ \underline{21} \quad \underline{21} \\ n = -3 \end{array} $	$ \begin{array}{r} 39 - 8n = -8(3 + 4n) + 3n \\ \underline{-8} \qquad \qquad \underline{-8} \\ 4\frac{7}{8} + n = 3 + 4n + 3n \\ 4\frac{7}{8} + n = 3 + 7n \\ \underline{-3} \qquad \underline{-3} \\ 1\frac{7}{8} + n = 7n \\ \underline{-n} \quad \underline{-n} \\ \underline{1\frac{7}{8}} = 6n \\ \underline{6} \quad \underline{6} \\ \frac{5}{16} = n \end{array} $

2. Describe the different first steps the students used in solving the equation.
Josephine subtracted $3n$ and then distributed the -8 before subtracting again. Jackie divided everything by -8 to avoid distributing it, then she subtracted 3, then n , and finally divided by 6.
3. Which student made an error?
Jackie
4. Write a note to the student who made an error explaining the mistake they made.
When multiplying or dividing on both sides of an equal sign, always double check to make sure every term is being multiplied or divided.
5. Is -7 a solution to the equation $32 + 7x - 4 = 3x$? Justify your answer mathematically and using words.
Yes, the left-hand side equals the right-hand side when $x = -7$.

$$\begin{array}{r}
 32 + 7x - 4 = 3x \\
 \underline{-7x} \quad \underline{-7x} \\
 28 \quad -4x \\
 \underline{-4} \quad \underline{-4} \\
 -7 = x
 \end{array}$$

$$\begin{array}{r}
 32 + 7(-7) - 4 = 3(-7) \\
 -21 = -21 \checkmark
 \end{array}$$

For questions 6–10, solve each equation and verify the solution using substitution.

6. $\frac{2m-5}{3} = 4m - 3$

$$3\left(\frac{2m-5}{3} = 4m - 3\right)$$

$$2m - 5 = 12m - 9$$

$$\frac{-2m}{-5} = \frac{-2m}{-5}$$

$$-5 = 10m - 9$$

$$\frac{+9}{+9} = \frac{+9}{+9}$$

$$\frac{4}{10} = \frac{10m}{10}$$

$$\frac{2}{5} = m$$

$$\frac{2}{5} = m$$

$$\frac{2\left(\frac{2}{5}\right) - 5}{3} = 4\left(\frac{2}{5}\right) - 3$$

$$\frac{-\frac{21}{5}}{3} = \frac{8}{5} - \frac{15}{5}$$

$$-\frac{7}{5} = -\frac{7}{5} \checkmark$$

7. $-11 - 2p = 6p + 5(p + 3)$

$$-11 - 2p = 6p + 5p + 15$$

$$\frac{+2p}{+2p} = \frac{+2p}{+2p}$$

$$-11 = 13p + 15$$

$$\frac{-15}{-15} = \frac{-15}{-15}$$

$$\frac{-26}{13} = \frac{13p}{13}$$

$$\frac{-2}{1} = p$$

$$-2 = p$$

$$-11 - 2(-2) = 6(-2) + 5((-2) + 3)$$

$$-11 + 4 = -12 - 10 + 15$$

$$-7 = -7 \checkmark$$

8. $\frac{2}{5} + 13k - 1 = 5k - \frac{3}{5}$

$$\frac{-5k}{-5k} = \frac{-5k}{-5k}$$

$$-\frac{3}{5} + 8k = -\frac{3}{5}$$

$$\frac{+3}{+3} = \frac{+3}{+3}$$

$$\frac{8k}{8} = \frac{0}{8}$$

$$\frac{k}{1} = \frac{0}{1}$$

$$k = 0$$

$$\frac{2}{5} + 13(0) - 1 = 5(0) - \frac{3}{5}$$

$$\frac{2}{5} - 1 = -\frac{3}{5}$$

$$-\frac{3}{5} = -\frac{3}{5} \checkmark$$

9. $3.1 + 6.8z - 2.4 = 0.3z - 4$

$$\frac{+4}{+4} = \frac{+4}{+4}$$

$$6.8z + 4.7 = 0.3z$$

$$\frac{-6.8z}{-6.8z} = \frac{-6.8z}{-6.8z}$$

$$\frac{4.7}{-6.5} = \frac{-6.5z}{-6.5}$$

$$-0.72308... = z \text{ or } z = -\frac{47}{65}$$

$$3.1 + 6.8\left(-\frac{47}{65}\right) - 2.4 = 0.3\left(-\frac{47}{65}\right) - 4$$

$$0.7 - \frac{1,598}{325} = -\frac{141}{650} - 4$$

$$-\frac{2,741}{650} = -\frac{2,741}{650} \checkmark \quad \text{or}$$

$$-4.21692... = -4.21692... \checkmark$$

$$10. \quad \frac{1}{2}x - 2(x + 1) = 3 + 1.5x + 1$$

$$\left(\frac{1}{2}x - 2(x + 1) = 3 + 1.5x + 1\right) 2$$

$$x - 4(x + 1) = 6 + 3x + 2$$

$$x - 4x - 4 = 3x + 8$$

$$\quad \quad \quad \begin{array}{r} +4 \qquad \qquad +4 \\ \hline \end{array}$$

$$-3x = 3x + 12$$

$$\begin{array}{r} -3x \quad -3x \\ \hline \end{array}$$

$$\begin{array}{r} -6x \quad 12 \\ \hline \end{array}$$

$$\begin{array}{r} -6 \quad -6 \\ \hline \end{array}$$

$$x = -2$$

$$\frac{1}{2}(-2) - 2(-2 + 1) = 3 + 1.5(-2) + 1$$

$$-1 + 2 = 4 - 3$$

$$1 = 1 \checkmark$$

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Lesson 2 Practice

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1. Solve the equation and verify the solution using substitution.

$$\begin{aligned}
 10\left[4 - \frac{1}{2}(2 + x) = \frac{2}{5}(-1 + 3x)\right] & \qquad 4 - \frac{1}{2}(2 + 2) = \frac{2}{5}(-1 + 3 \cdot 2) \\
 40 - 5(2 + x) = 4(-1 + 3x) & \qquad 4 - \frac{1}{2}(4) = \frac{2}{5}(-1 + 6) \\
 40 - 10 - 5x = -4 + 12x & \qquad 4 - 2 = \frac{2}{5}(5) \\
 30 - 5x = -4 + 12x & \qquad 2 = 2\checkmark \\
 34 = 17x & \\
 x = 2 &
 \end{aligned}$$

The point-slope form of a line is given by the formula $y - y_1 = m(x - x_1)$.

2. Use the distributive property to rewrite the point-slope formula without parentheses.

$$y - y_1 = mx - mx_1$$

3. Use your answer from question 2 to rearrange the point-slope formula to solve for the variable y .

$$y - y_1 = mx - mx_1$$

$$\underline{+y_1} \quad +y_1$$

$$y = mx - mx_1 + y_1$$

4. Compare your result in question 3 to the slope-intercept form of a line. Using your answer from question 3 what is the y -intercept in your equation?

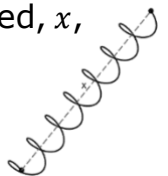
$$y_1$$

Rearrange the formulas to isolate the specified variables.

5. Hooke's Law, $F = -kx$, is used in physics to find the force, F , needed to stretch or compress a spring, based on the length the spring is stretched or compressed, x , and the spring constant, k . Solve the equation for x .

$$\frac{F}{-k} = \frac{-kx}{-k}$$

$$x = -\frac{F}{k}$$



6. The formula for the volume of a sphere is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Solve the equation for the radius, r .

$$\frac{3}{4}\left[V = \frac{4}{3}\pi r^3\right]$$

$$\frac{\frac{3}{4}V}{\pi} = \frac{\pi r^3}{\pi}$$

$$\sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{r^3}$$

$$r = \sqrt[3]{\frac{3V}{4\pi}}$$

7. The surface area of a rectangular prism is given by the formula $SA = 2(lw + lh + hw)$. Rearrange the formula to solve for the width, w .

$$SA = 2(lw + lh + hw)$$

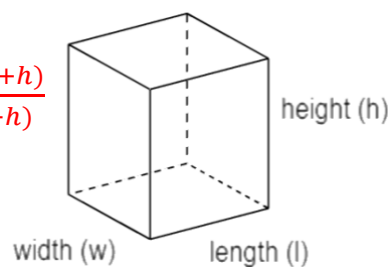
$$SA = 2lw + 2lh + 2hw$$

$$\frac{-2lh}{2(l+h)} \quad \frac{-2lh}{2(l+h)}$$

$$SA - 2lh = 2lw + 2hw$$

$$\frac{SA - 2lh}{2(l+h)} = \frac{2w(l+h)}{2(l+h)}$$

$$\frac{SA - 2lh}{2(l+h)} = w$$



The area of a composite shape can be found using several methods. One method is to divide the total area into sub-problems, then find the area of each one, and add them together to get the total area. Another method is to calculate the area of a larger, simpler figure and subtract off the unnecessary pieces. Daeun and Jose have been asked to find the area of the figure below. Daeun chooses to use sub-problems while Jose uses the subtraction method. Their work is shown below. Use to their work to answer the questions that follow.

Daeun's Work	Jose's Work
The area is equal to sum of the inside areas: - rectangle: $a \times b$ - rectangle: $a \times (d + e)$ - rectangle: $c \times d$ - triangle: $\frac{1}{2}e \times c$	The area is equal to the area of the large rectangle, minus the extra pieces: - large rectangle: $(a + c) \times (b + d + e)$ - extra rectangle: $b \times c$ - extra triangle: $\frac{1}{2}e \times c$
Total Area: $A = ab + cd + \frac{1}{2}ec + a(d + e)$	Total Area: $A = (a + c)(b + d + e) - bc - \frac{1}{2}ec$

8. Use the known lengths to label the lengths of the dashed lines (a, b, c, d, e , etc).
9. Simplify both equations for the total area so that there are no more parentheses in either one.

$$A = ab + cd + \frac{1}{2}ec + ad + ae$$

$$A = ab + ad + ae + cd + \frac{1}{2}ce$$

10. Did Daeun and Jose get the same answer for the total area? If so, rearrange Daeun's simplified equation so that it is the same as Jose's. If not, identify and correct any mistakes you find.

$$\text{Yes } A = ab + cd + \frac{1}{2}ec + ad + ae = ab + ad + ae + cd + \frac{1}{2}ce$$

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Lesson 3 Practice

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Choose numbers from the positive digits 1 – 9 to fill in the boxes below for questions 1–4. *You may only use each digit one time.*

$$y = \frac{\boxed{3}}{\boxed{4}} \cdot x + \boxed{5}$$

1. List the variables in your equation.

x, y

2. List the parameters in your equation.

Answers will vary.

$\frac{3}{4}$

3. Choose new values for the parameters so that the value of y will be as **large** as possible if $x = 1$. *You will still use the digits 1 – 9 only one time, but you may re-use numbers from the previous equation.* How do you know your parameters make y as large as possible?

$$y = \frac{8}{1}x + 9$$

9 is the largest starting point and $\frac{8}{1}$ is largest slope without repeating digits.

4. Choose new values for the parameters so that the value of y will be as **small** as possible if $x = 1$. *You will still use the digits 1 – 9 only one time, but you may re-use numbers from the previous equation.* How do you know your parameters make y as small as possible?

$$y = \frac{2}{9}x + 1$$

1 is the smallest starting point and $\frac{2}{9}$ is smallest slope without repeating digits.

5. Xander looked at the equation $3y + 4 = 9x - 11$ and said, "The variables are x and y . The parameters are 3, 4, 9, and -11 ." Z'Kera said, "I agree about the variables, but the parameters are 3 and 5." Who is correct, Xander, Z'Kera, or both?

Neither. When the equation is put into slope-intercept form it is $y = 3x - 5$. The slope/parameter is 3.

Yvonne earns a fixed amount of money per day as a substitute teacher and earns an hourly rate for each hour that she works as an after-school tutor each week. Yvonne's pay can be modeled by the function $f(x) = 15x + 100$.

6. Which value(s) in the equation are variables?

$f(x)$ and x

7. Which value(s) in the equation are parameters?

15

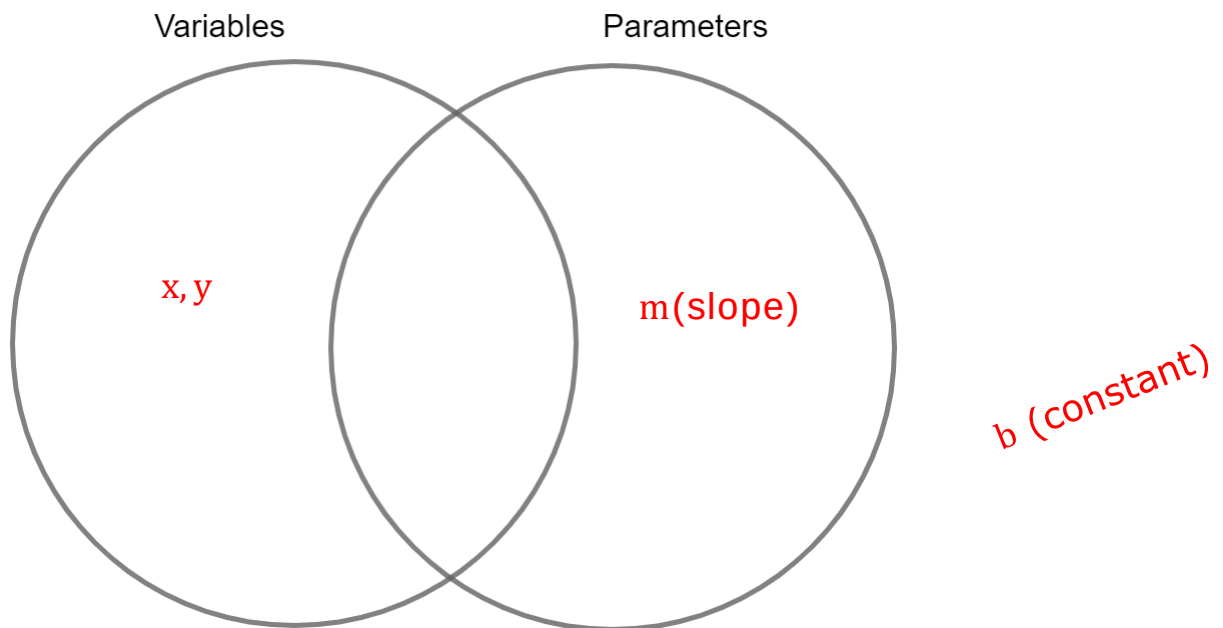
8. What does the value 15 mean in terms of Yvonne's pay?

Yvonne makes \$15 per hour.

9. What does the value 100 mean in terms of Yvonne's pay?

It is the fixed pay rate for substitute teaching.

10. Complete the Venn Diagram to compare and contrast variables and parameters.



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Lesson 4 Practice

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Marquise wants to save up for a pair of new sneakers that cost \$120. He has already saved \$37.50. He downloaded an app that will allow him to add ten cents to his savings account whenever he pays with his debit card. He uses his debit card to make an average of three purchases per day.

1. Define the variables in this context.

x = number of days

y = total saved

2. What is the parameter?

He earns an average of \$0.30 per day.

3. What is the constant?

37.50

4. Which variable represents the output in this situation?

y

5. Complete the table to write an equation that represents the context.

Total saved	=	# purchases	·	# days	+	Starting balance
y	=	0.30	·	x	+	37.50

6. Solve the equation for the unknown.

$$120 = 0.3x + 37.50$$

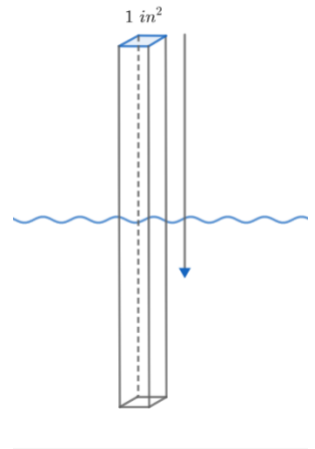
$$82.50 = 0.3x$$

$$x = 275$$

7. Interpret the solution.

It will take 275 days to save the money.

To measure air pressure, scientists consider a 1 in^2 column of air that reaches from sea level all the way up to the atmosphere. Even though air seems “empty”, it actually has weight, which exerts force on objects on the ground! The air applies 14.7 pounds of pressure per square inch (14.7 psi) at sea level.



Water weighs much more than air. The deeper you dive below the surface, the pressure the water exerts. For every foot of depth, the water weighs an additional 0.445 lbs beyond the weight of the 1 in^2 column of air.

8. Define the variables in this context.

$x = \text{depth}$

$y = \text{weight}$

9. Write an equation that can be used to find the pressure exerted on an object in terms of the depth below sea level.

$$y = 0.445x + 14.7$$

10. The human body can take 58 psi of pressure. Use your equation to find the maximum depth under water a human could survive without being crushed by the pressure of the water.

$$58 = 0.445x + 14.7$$

$$43.3 = 0.445x$$

$$x \approx 97.3 \text{ feet}$$

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Jayne and Tolu are each raising money for charity for their club at school. They decide to make it a friendly competition. Jayne's club is hosting a walk-a-thon. She collects a total of \$35 for every mile she walks. Tolu's club is holding a carwash. He earns \$25 for charity for every car he washes, and one customer gave him a \$100 donation and did not even get a car wash!

1. Define a variable to represent the unknown.
 $x = \text{number of completed tasks}$
2. Write an expression for the amount of money Jayne raised for her club.
 $35x$
3. Write an expression for the amount of money Tolu raised for his club.
 $25x + 100$
4. When the events are over, they realized that they both completed the same number of tasks and earned the same amount of money. Set the two expressions equal to each other to create an equation.
 $35x = 25x + 100$
 $10x = 100$
5. Solve the equation and interpret your answer.
 $10x = 100$
 $x = 10$
They completed ten tasks each.
6. How much money Jayne and Tolu each raise?
\$350 each
7. If Jayne earned \$37 per mile instead, would it have been possible for the two friends to earn the same amount of money for the same number of tasks?
 $37x = 25x + 100$
 $12x = 100$
 $x = \frac{25}{3}$ or $8\frac{1}{3}$
No, because you can't have a fractional task.

Tran is hungry for lunch, but he is on a budget. Two local Mexican restaurants have affordable lunch specials. Leo Tapatio charges \$2 per taco and \$5 for a plate of rice and beans and a drink. Taco Loco charges \$1 per small street-taco and has an all you can eat salad and salsa bar for \$7.50.

8. Define a variable to represent the unknown.

x = number of tacos

Tran wants to spend the same amount of money at each restaurant for the same number of tacos, so created and solved an equation. His work is shown below.

$$\begin{aligned}2x + 5 &= x + 7.5 \\2x + 5 - x &= x + 7.5 - x \\x + 5 &= 7.5 \\x + 5 - 5 &= 7.5 - 5 \\x &= 2.5\end{aligned}$$

9. Interpret Tran's answer.

2.5 tacos would cost the same at either restaurant.

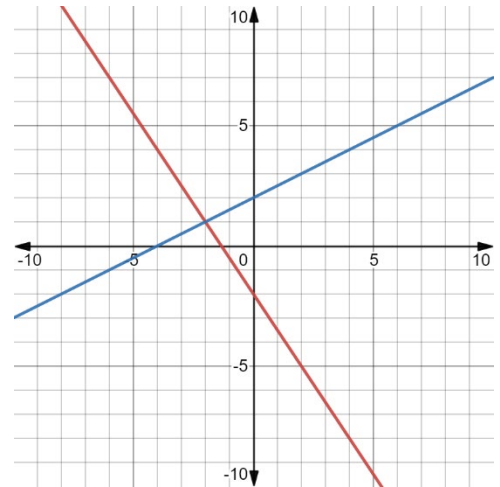
10. Explain whether Tran's solution to the equation is viable in this context.

No, because you can't buy half a taco.

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Lesson 6 Practice

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The graph of the system of equations $\begin{cases} x - 2y = -4 \\ -3x - 2y = 4 \end{cases}$ is shown.



1. To determine the solution to a system of equations, identify the

- ☒ point of intersection.
- ☐ slope.
- ☐ y-intercept.

2. Determine the solution to the system of equations from the graph.

$(-2, 1)$

Verify algebraically that that your solution satisfies both equations in the system.

3. $x - 2y = -4$
 $-2 - 2(1) \stackrel{?}{=} -4$
 $-2 - 2 = -4 \checkmark$

4. $-3x - 2y = 4$
 $-3(-2) - 2(1) \stackrel{?}{=} 4$
 $6 - 2 = 4 \checkmark$

5. Choose all of the systems of equations that are equivalent to the original system of equations, $\begin{cases} x - 2y = -4 \\ -3x - 2y = 4 \end{cases}$. Explain how you know the systems are equivalent.

Multiply the first equation by 4.

$\begin{cases} y = \frac{1}{2}x + 2 \\ y = -\frac{3}{2}x - 2 \end{cases}$	$\begin{cases} y = -\frac{1}{2}x + 2 \\ y = -\frac{3}{2}x - 2 \end{cases}$	$\begin{cases} 4x - 8y = -16 \\ -3x - 2y = 4 \end{cases}$	$\begin{cases} x - 2y = -4 \\ -x - \frac{2}{3}y = \frac{4}{3} \end{cases}$
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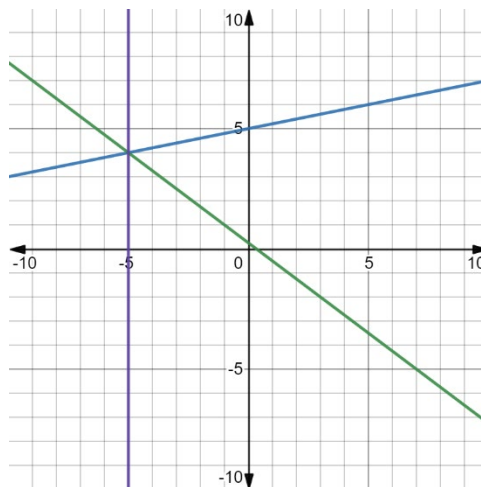
Write both equations in slope-intercept form by solving for y.

Multiply the second equation by $\frac{1}{3}$.

6. Verify your choices in question 5 algebraically.

$x - 2y = -4 \rightarrow$ $-\frac{1}{2}(-2y = -x - 4) \rightarrow$ $y = \frac{1}{2}x + 2 \checkmark$ $-3x - 2y = 4 \rightarrow$ $-\frac{1}{2}(-2y = 3x + 4) \rightarrow$ $y = -\frac{3}{2}x - 2 \checkmark$	$x - 2y = -4 \rightarrow$ $-\frac{1}{2}(-2y = -x - 4) \rightarrow$ $y = \frac{1}{2}x + 2 \boxtimes$	$4(x - 2y = -4) \rightarrow$ $4x - 8y = -16 \checkmark$	$\frac{1}{3}(-3x - 2y = 4) \rightarrow$ $-x - \frac{2}{3}y = \frac{4}{3} \checkmark$
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The graphs of the system of equations $\begin{cases} y = \frac{1}{5}x + 5 \\ y = -\frac{3}{4}x + \frac{1}{4} \end{cases}$ and $\begin{cases} x = -5 \\ y = \frac{1}{5}x + 5 \end{cases}$ are shown.



7. The solution to the system $\begin{cases} y = \frac{1}{5}x + 5 \\ y = -\frac{3}{4}x + \frac{1}{4} \end{cases}$ is $(-5, 4)$.

8. The solution to the system $\begin{cases} x = -5 \\ y = \frac{1}{5}x + 5 \end{cases}$ is $(-5, 4)$.

9. Fill in the blanks with words from the word bank to complete the sentence.

The two systems of equations are <u>equivalent</u> because the two systems result in <u>the same</u> solution(s).	Word Bank: ○ equivalent ○ equal ○ not equivalent ○ not equal ○ the same ○ different
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10. Write a third system of equations that is equivalent, but **not** equal to the system

of equations $\begin{cases} y = \frac{1}{5}x + 5 \\ y = -\frac{3}{4}x + \frac{1}{4} \end{cases}$

Answers will vary.

$$\begin{cases} y = \frac{1}{5}x + 5 \\ 3x + 4y = 1 \end{cases}$$

$$\begin{aligned} &\left(y = -\frac{3}{4}x + \frac{1}{4}\right) 4 \\ &4y = -3x + 1 \\ &\underline{+3x + 3x} \\ &3x + 4y = 1 \end{aligned}$$

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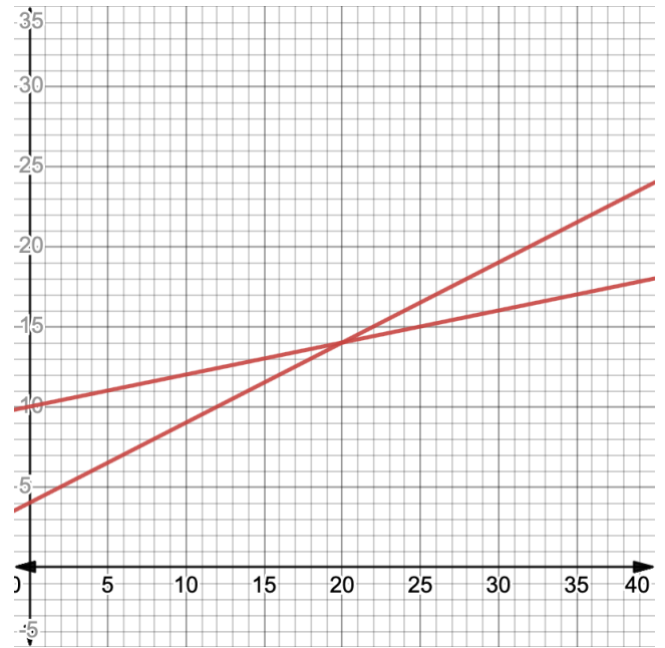
Use the system of equations $\begin{cases} y = 0.5x + 4 \\ y = 0.2x + 10 \end{cases}$ to complete the following.

$$\frac{2}{10} = \frac{1}{5}$$

- Graph the system of equations on the coordinate plane.

- Complete the statement.

The point (20 , 14) is the solution to the system.



Alexis has a total of 14 nickels and dimes worth \$1.25. The system of equations $\begin{cases} n = 14 - d \\ n = 25 - 2d \end{cases}$ can be used to represent the amount of quarters, q , and the amount of dimes, d .

The graph of the system is shown.

- Label the axes.
- What are any possible constraints on the x -values?

$$x \geq 0$$

- What are any possible constraints on the y -values?

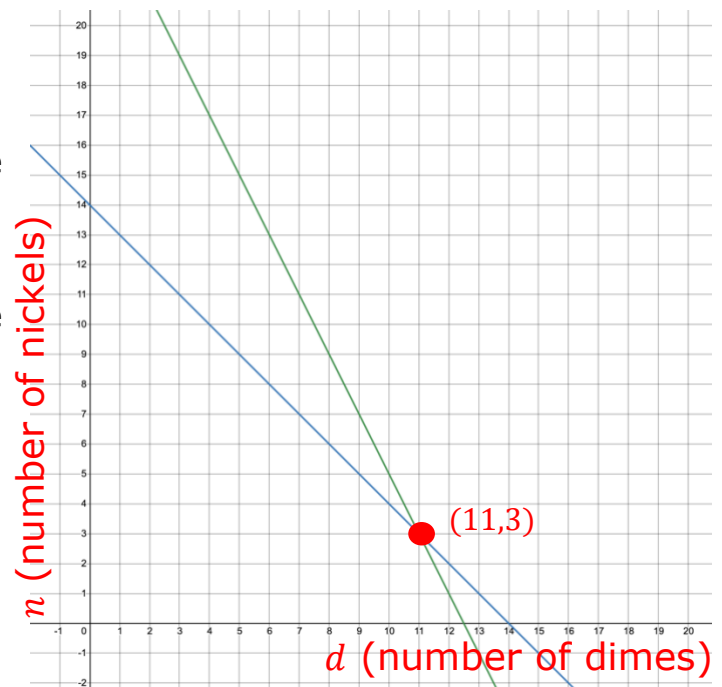
$$y \geq 0$$

- What is the solution to this system of equations?

(11,3)

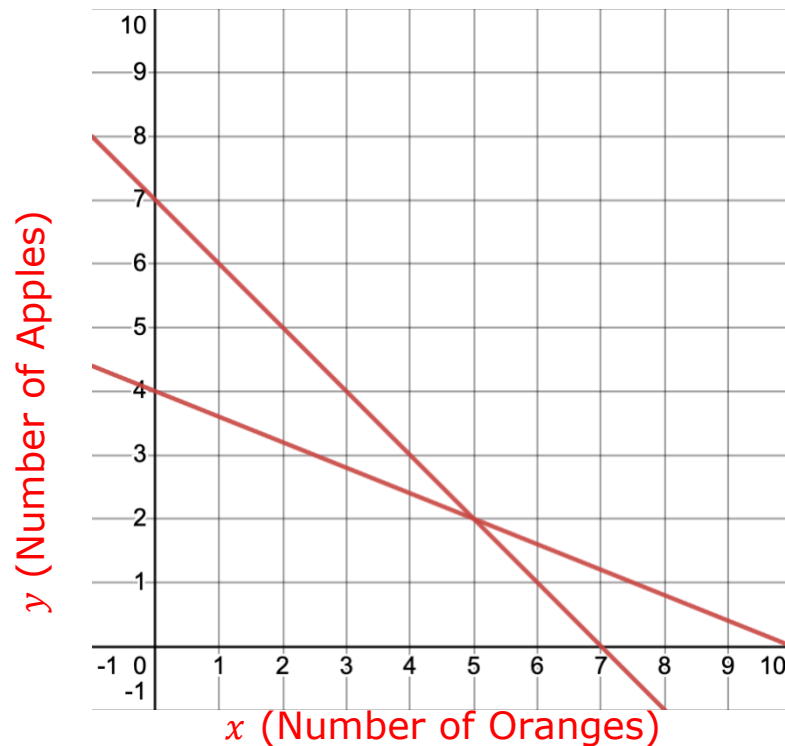
- Interpret the meaning of the solution in the context.

Alexis has 11 dimes and 3 nickels, totaling \$1.25.



Chen went to the grocery store and bought apples for \$1.50 each and oranges for \$0.60 each. He bought a total of 7 pieces of fruit and spent \$6. The system of equations $\begin{cases} y = 7 - x \\ y = -\frac{2}{5}x + 4 \end{cases}$ can be used to represent the situation where x is the number of oranges and y is the number of apples bought.

8. Graph the system of equations. Label each line with the fruit it represents.



9. What is the solution to the system?

(5,2)

10. Interpret the meaning of the solution in the context.

Chen bought 5 oranges and 2 apples at the grocery store.

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Lesson 8 Practice

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For questions 1–2, solve each system of equations using substitution.

1. $\begin{cases} y = 9x - 4 \\ y = 6x + 2 \end{cases}$

$$\begin{aligned} 6x + 2 &= 9x - 4 \\ 2 &= 3x - 4 \\ 6 &= 3x \\ 2 &= x \end{aligned}$$
$$\begin{aligned} y &= 9(2) - 4 \\ y &= 18 - 4 \\ y &= 14 \end{aligned}$$

(2,14)

2. $\begin{cases} 3x - y = 4 \\ y = \frac{1}{3}x + 4 \end{cases}$

$$\begin{aligned} 3x - \left(\frac{1}{3}x + 4\right) &= 4 \\ 3x - \frac{1}{3}x - 4 &= 4 \\ \frac{8}{3}x - 4 &= 4 \\ \frac{8}{3}x &= 8 \\ x &= 3 \end{aligned}$$
$$\begin{aligned} y &= \frac{1}{3}(3) + 4 \\ y &= 1 + 4 \\ y &= 5 \end{aligned}$$

(3,5)

Alexander was looking for new jobs. He got two different job offers. Job A offers Alexander an hourly position making \$18 an hour plus a bonus of \$750 a year. Job B offers Alexander an annual salary of \$47,000. The system of equations $\begin{cases} s = 18h + 750 \\ s = 47,000 \end{cases}$ represents the situation, where s is the salary Alexander will earn and h is the number of hours worked.

3. Solve the system of equations using substitution.

$$\begin{aligned} 47,000 &= 18h + 750 \\ 46,250 &= 18h \\ 2,569.\bar{4} &= h \end{aligned}$$

(2569. $\bar{4}$, 47000)

4. Interpret the solution to the system.
If Alexander works 2,569. $\bar{4}$ hours at Job A, he will make the same salary as Job B, \$47,000.

Yazmin is using substitution to solve the system of equations $\begin{cases} y = 7x + 10 \\ x = -y + 2 \end{cases}$. Yazmin's partial work is shown below.

$$\begin{cases} y = 7x + 10 \\ x = -y + 2 \end{cases}$$

$$\begin{aligned} y &= 7(-y + 2) + 10 \\ y &= -7y + 2 + 10 \\ y &= -7y + 12 \\ 8y &= 12 \\ y &= \frac{12}{8} \end{aligned}$$

5. Yazmin made an error when solving for y . Circle the step where the mistake was made.

6. Find the correct solution for y .

$$\begin{aligned} y &= -7y + 14 + 10 \\ y &= -7y + 24 \\ 8y &= 24 \\ y &= 3 \end{aligned}$$

7. Using the corrected y - value, find the solution to the system.

$$\begin{aligned} x &= -3 + 2 & 3 &= 7x + 10 \\ x &= -1 & \text{or} & -7 = 7x \\ & & & -1 = x \end{aligned} \quad (-1, 3)$$

8. Yazmin substituted the second equation into the first to find y . Solve the system by substituting the first equation into the second to find x .

$$\begin{aligned} x &= -(7x + 10) + 2 & y &= 7(-1) + 10 \\ x &= -7x - 10 + 2 & y &= -7 + 10 \\ 8x &= -8 & y &= 3 \\ x &= -1 & & \end{aligned} \quad (-1, 3)$$

9. Did you get the same solution using both methods?

Yes

10. Solve the system of equations $\begin{cases} 2x - 4y = 8 \\ y = -\frac{3}{4}x + 3 \end{cases}$ by using substitution.

$$\begin{aligned} 2x - 4\left(-\frac{3}{4}x + 3\right) &= 8 & y &= -\frac{3}{4}(4) + 3 \\ 2x + 3x - 12 &= 8 & y &= -3 + 3 \\ 5x - 12 &= 8 & & \\ 5x &= 20 & x &= 4 \end{aligned} \quad (4, 0)$$

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Use the system of equations $\begin{cases} x + 4y = 3 \\ 0.5x - 4y = 6 \end{cases}$ to complete questions 1 – 4.

1. Complete the statement.

- ☒ Adding
☐ Subtracting

the original equations in this system would result in eliminating one of the variables,

- ☐ x .
☒ y .

2. Perform the selected operation selected in question 1.

$$\begin{array}{r} x + 4y = 3 \\ +0.5x - 4y = 6 \\ \hline 1.5x = 9 \end{array}$$

3. Solve the system using your equation from question 2.

$$1.5x = 9$$

$$x = 6$$

$$x + 4y = 3$$

$$6 + 4y = 3$$

$$4y = -3$$

$$y = -\frac{3}{4}$$

$$\left(6, -\frac{3}{4}\right)$$

or

$$(6, -0.75)$$

4. Verify the solution algebraically using substitution.

$$6 + 4\left(-\frac{3}{4}\right) = 6 - 3 = 3 \checkmark$$

$$0.5(6) - 4\left(-\frac{3}{4}\right) = 3 + 3 = 6 \checkmark$$

5. Solve the system of equations using elimination.

$$\begin{cases} 5p + 2h = 13 \\ (3p - h = -12)2 \end{cases}$$

$$\begin{array}{r} 5p + 2h = 13 \\ +6p - 2h = -24 \\ \hline 11p = -11 \\ p = -1 \end{array}$$

$$\begin{array}{r} 5(-1) + 2h = 13 \\ -5 + 2h = 13 \end{array}$$

$$\begin{array}{r} 2h = 18 \\ h = 9 \end{array}$$

$$(-1, 9)$$

6. Verify the solution from question 5 algebraically using substitution.

$$5(-1) + 2(9) = -5 + 18 = 13 \checkmark$$

$$3(-1) - 9 = -3 - 9 = -12 \checkmark$$

7. Solve the system of equations algebraically.

$$\begin{array}{lcl}
 \begin{cases} 3x + y = 5 \\ 5x - 4y = -12 \end{cases} & \begin{array}{l} 12x + 4y = 20 \\ +5x - 4y = -12 \\ \hline 17x = 8 \\ x = \frac{8}{17} \end{array} & \begin{array}{l} 3\left(\frac{8}{17}\right) + y = 5 \\ \frac{24}{17} + y = 5 \\ y = \frac{85}{17} - \frac{24}{17} \\ y = \frac{61}{17} \end{array}
 \end{array}
 \quad \left(\frac{8}{17}, \frac{61}{17}\right)$$

8. Verify the solution from question 7 algebraically using substitution.

$$\begin{aligned}
 3\left(\frac{8}{17}\right) + \left(\frac{61}{17}\right) &= \frac{24}{17} + \frac{61}{17} = \frac{85}{17} = 5 \quad \checkmark \\
 5\left(\frac{8}{17}\right) - 4\left(\frac{61}{17}\right) &= \frac{40}{17} - \frac{244}{17} = -\frac{204}{17} = -12 \quad \checkmark
 \end{aligned}$$

9. Mia is solving the system of equations $\begin{cases} 2x + 3y = 7 \\ 3x - y = -6 \end{cases}$ using elimination. In order to eliminate the x , she multiplies the first equation by a factor of 3 and the second equation by a factor of 2. Explain whether her method will work.

Yes, the x terms will be eliminated by subtracting the equations.

$$\begin{aligned}
 3(2x + 3y = 7) &\rightarrow 6x + 9y = 21 \\
 2(3x - y = -6) &\rightarrow 6x - 2y = -12
 \end{aligned}$$

10. Mia's system would be simpler to solve by eliminating the y . Describe why this is true and then solve Mia's system.

Only the second equation needs to be manipulated to eliminate the y .

$$3(3x - y = -6) \rightarrow 9x - 3y = -18$$

$$\begin{array}{lcl}
 \begin{array}{l} 2x + 3y = 7 \\ +9x - 3y = -18 \\ \hline 11x = -11 \\ x = -1 \end{array} & \begin{array}{l} 2(-1) + 3y = 7 \\ -2 + 3y = 7 \\ 3y = 9 \\ y = 3 \end{array} & (-1, 3)
 \end{array}$$

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1. Solve the system of equations using elimination.

$$\begin{array}{rcl} \begin{cases} 2x + 3y = 9 \\ \left(\frac{1}{2}x + \frac{2}{3}y = 4\right) - 4 \end{cases} & \begin{array}{r} 2x + 3y = 9 \\ -2x - \frac{8}{3}y = -16 \\ \hline \frac{1}{3}y = -7 \\ y = -21 \end{array} & \begin{array}{r} 2x + 3(-21) = 9 \\ 2x - 63 = 9 \\ 2x = 72 \\ x = 36 \end{array} \end{array} \quad (36, -21)$$

2. Verify the solution to question 1 algebraically using substitution.

$$\begin{aligned} 2(36) + 3(-21) &= 72 - 63 = 9 \checkmark \\ \frac{1}{2}(36) + \frac{2}{3}(-21) &= 18 - 14 = 4 \checkmark \end{aligned}$$

Pam has two part time jobs. At one job, she works as a cashier and makes \$8 per hour. At the second job, she works as a tutor and makes \$12 per hour. One week she worked 30 hours and made \$268. This situation can be represented by the system of equations, $\begin{cases} 8c + 12t = 268 \\ c + t = 30 \end{cases}$, where c is the number of hours she worked as a cashier and t is the number of hours she worked as a tutor.

3. The system could be solved using both substitution or elimination. Which method do you prefer to use?

Elimination

4. Solve the system of equations using your preferred method.

$$\begin{array}{rcl} -8(c + t = 30) & \rightarrow & -8c - 8t = -240 \\ \begin{array}{r} 8c + 12t = 268 \\ + -8c - 8t = -240 \\ \hline 4t = 28 \\ t = 7 \end{array} & \begin{array}{r} c + 7 = 30 \\ c = 23 \end{array} & (23, 7) \end{array}$$

5. Interpret your solution.

Pam worked 23 hours as a cashier and 7 hours as a tutor.

6. Explain whether negative values for c and t are viable in this context.

No, it does not make sense to work a negative number of hours.

The Johnson and Green families went to a fast-food restaurant for lunch. The Johnson's bought 4 cheeseburgers and 3 medium fries for \$16.53. The Greens bought 5 cheeseburgers and 4 medium fries for \$21.11. This situation can be represented by the system of equations shown where c represents the cost of a cheeseburger and f represents the cost of medium fries.

$$\begin{aligned} (4c + 3f = 16.53) - 4 &\rightarrow -16c - 12f = -66.12 \\ (5c + 4f = 21.11) - 3 &\rightarrow 15c + 12f = 63.33 \end{aligned}$$

7. Explain why substitution is not an efficient method to solving the system.

It would take more than one step to isolate a variable and there would be a lot of decimals.

8. Solve the system of equations.

$$\begin{array}{rcl} -16c - 12f & = & -66.12 \\ +15c + 12f & = & 63.33 \\ \hline -c & = & -2.79 \\ c & = & 2.79 \end{array} \qquad \begin{array}{rcl} 4(2.79) + 3f & = & 16.53 \\ 11.16 + 3f & = & 16.53 \\ 3f & = & 5.37 \\ f & = & 1.79 \end{array} \qquad (2.79, 1.79)$$

9. Interpret the solution.

The cost of a cheeseburger is \$2.79 and the cost of medium fries is \$1.79.

10. Determine the constraints on c and f in this context.

(It does not make sense for cost to be negative)

$$\begin{aligned} 0 &\leq c \leq 25 \\ 0 &\leq f \leq 25 \end{aligned}$$

(It also does not make sense for the cost of a burger/fries at a fast-food restaurant to be too high)

Tyler owns a food truck that sells chicken wings and hamburger sliders. Each order of chicken wings sells for \$7.25 while each order of hamburger sliders sells for \$3. On Friday, Tyler made a total of \$595 in revenue from all sliders and wings sales. There were twice as many chicken wing orders sold than there were sliders sold.

1. Define the variables to write the system of equations.

Let c = chicken wing orders sold

Let s = sliders sold

2. Write a system of equations to determine the number of chicken wing orders sold and the number of hamburger sliders sold.

$$\begin{cases} 7.25c + 3s = 595 \\ c = 2s \end{cases}$$

3. Solve the system using the method of your choice.

$$\begin{aligned} 7.25(2s) + 3s &= 595 & c &= 2(34) \\ 14.5s + 3s &= 595 & c &= 68 \\ 17.5s &= 595 \\ s &= 34 \end{aligned}$$

4. The number of chicken wing orders on Friday was 68. The number of hamburger slider orders was 34.

Logan has \$0.87 worth of pennies and nickels. He has a total of 31 coins altogether. Determine the number of pennies and nickels Logan has.

5. Write a system of equations that could be used to determine the number of pennies and nickels Logan has.

$$\begin{cases} 0.01p + 0.05n = 0.87 \\ p + n = 31 \end{cases}$$

6. Solve the system of equations.

$$\begin{aligned} p &= 31 - n \\ 0.01(31 - n) + 0.05n &= 0.87 & p + n &= 31 \\ 0.31 - 0.01n + 0.05n &= 0.87 & p + 14 &= 31 \\ 0.04n &= 0.56 & p &= 17 \\ n &= 14 \\ 17 \text{ pennies} \\ 14 \text{ nickels} \end{aligned}$$

Axel wanted to make salsa, so he went to the grocery store to buy 7 tomatoes and 4 onions. He spent \$17.75. He didn't buy enough, so he went to the same store and bought 5 more tomatoes and 2 more onions, spending \$10.75.

7. Write a system of equations that could be used to find the price of tomatoes and onions.

$$\begin{cases} 7t + 4x = 17.75 \\ 5t + 2x = 10.75 \end{cases}$$

8. Solve the system of equations.

$$\begin{array}{rcl} -2(5t + 2x = 10.75) & \rightarrow & -10t - 4x = -21.5 \\ 7t + 4x = 17.75 & & 7(1.25) + 4x = 17.75 \\ \hline -10t - 4x = -21.5 & & 8.75 + 4x = 17.75 \\ -3t = -3.75 & & 4x = 9 \\ t = 1.25 & & x = 2.25 \end{array}$$

Tomatoes: \$1.25

Onions: \$2.25

Pressley is working two part-time jobs. He makes \$18 per hour lifeguarding and \$12 per hour as a cashier at the local grocery store. During spring break from school, he worked twice as many hours lifeguarding as he did at the local grocery store and earned \$144. How many hours did he work at each job?

9. Write a system of equations that could be used to determine the number of hours Pressley worked.

$$\begin{cases} 18x + 12c = 144 \\ x = 2c \end{cases}$$

10. Solve the system using the method of your choice.

$$\begin{array}{rcl} 18(2c) + 12c = 144 & & x = 2(3) \\ 36c + 12c = 144 & & x = 6 \\ 48c = 144 & & \\ c = 3 & & \end{array}$$

6 hours lifeguarding
3 hours as a cashier